Door Status Indicator Module

PCB DESIGN & HARDWARE ENGINEERING REPORT

Student Name: Yuvang Chikara
Assignment: KiCad PCB Design Module
Input Voltage: 5.0 V DC (Regulated)
Status: Validated & Production Ready

1. CIRCUIT OVERVIEW & OPERATING PRINCIPLE

The Door Status Indicator Module provides automated visual and acoustic feedback for access boundaries. When a door is closed, a magnetic field engages the SPDT reed switch (SW1), completing the circuit through current-limiting resistor R2 to illuminate the **Green LED (D2)**. When the door opens, the magnet shifts away, releasing the switch contact to the Normally Closed (NC) position. This energizes the **Red LED (D1)** via resistor R1 while simultaneously triggering the **5V Active Buzzer (BZ1)** for intrusion notification.

2. DELIVERABLE 3: COMPLETE KICAD SCHEMATIC CAPTURE

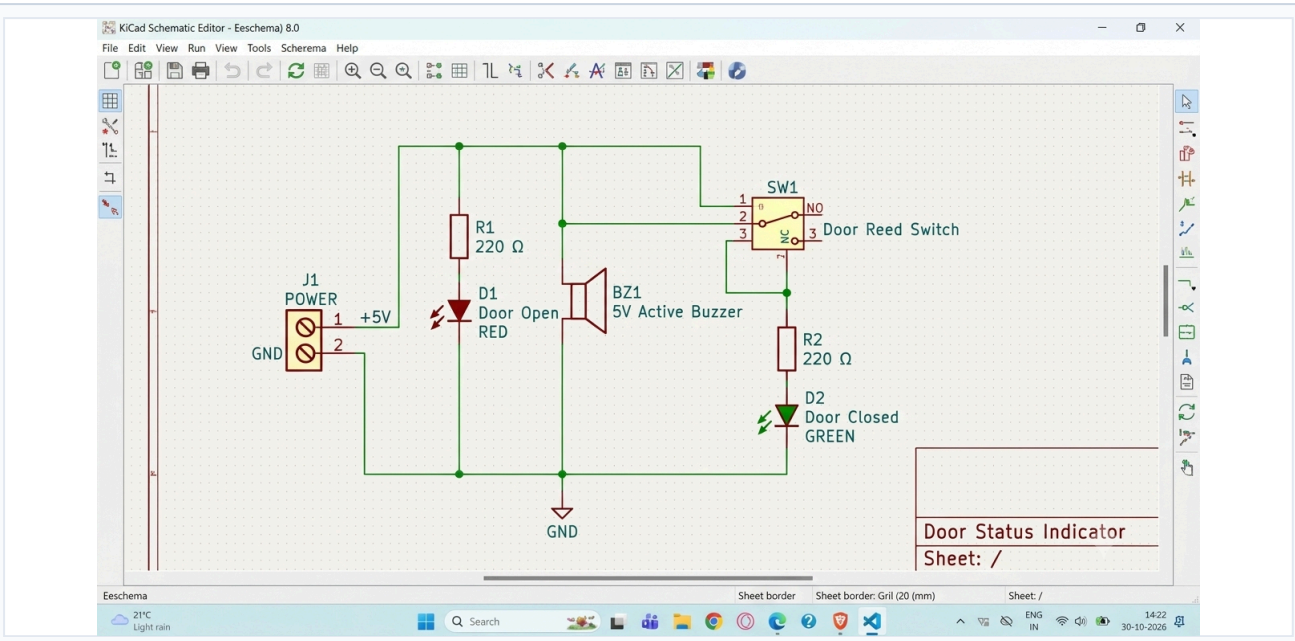


Figure 1.1: Completed Electrical Schematic Diagram in KiCad 8.0 (Eeschema) with standard symbol annotations.

3. CIRCUIT DESIGN EQUATIONS & ELECTRICAL ANALYSIS

LED Current Limiting Calculation

For standard 5mm indicator LEDs with a forward voltage of $V_f \approx 2.0$ V and a nominal supply $V_{CC} = 5.0$ V:

$$I_{LED} = (V_{CC} - V_f) / R = (5.0V - 2.0V) / 220\ \Omega = 13.6\ mA$$

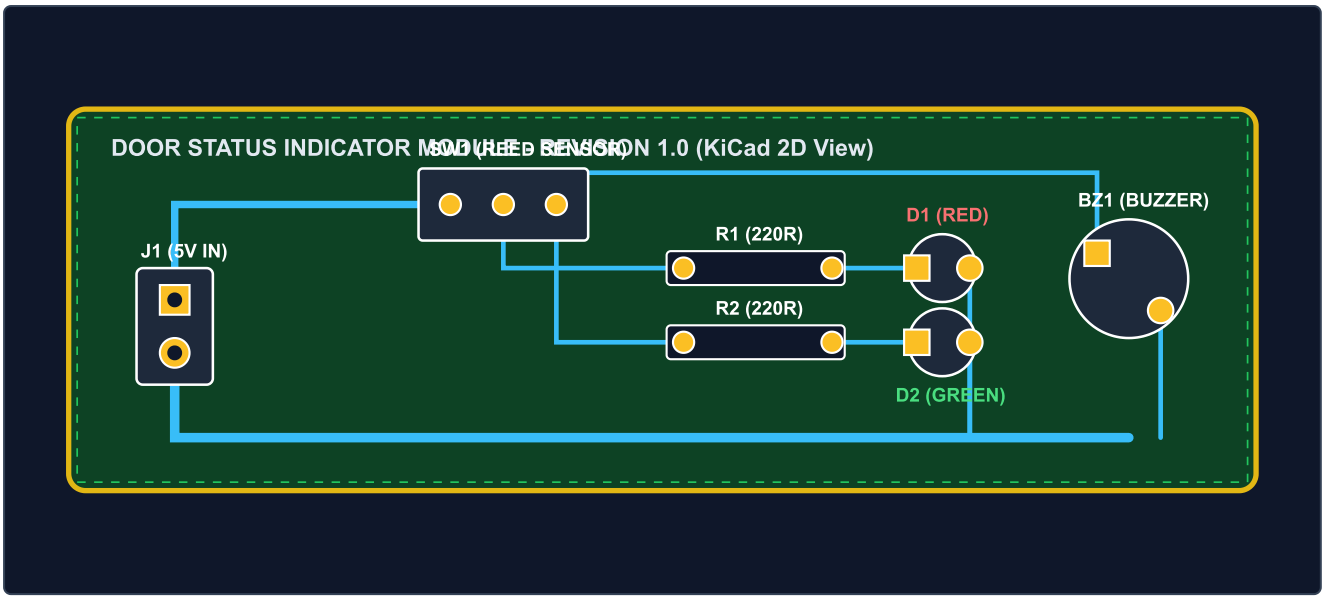
This ensures high visibility while operating well below the 20 mA continuous threshold.

Active Buzzer Circuit Architecture

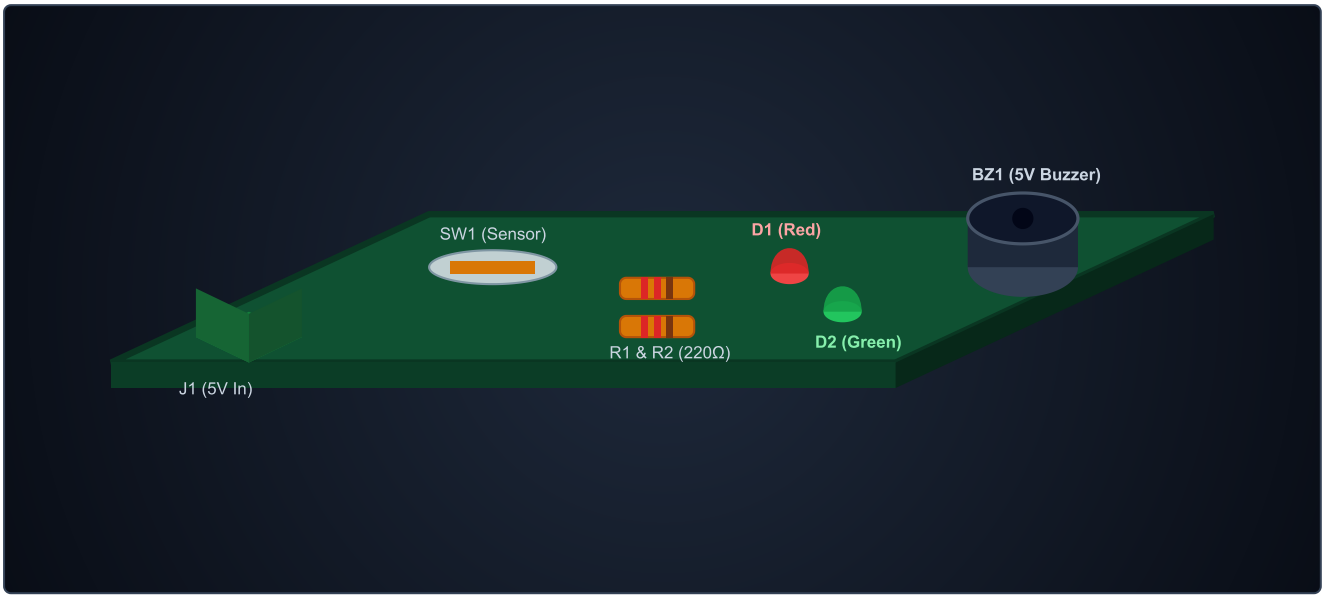
The 5V active piezo buzzer (BZ1) features internal oscillator circuitry requiring exactly 5V DC for actuation. It is placed in parallel across the door-open state net (SW1 Pin 2 to GND), receiving full rail voltage independently of R1's current-limiting path.

4. DELIVERABLES 4 & 5: ROUTED PCB FOOTPRINT & 2D LAYOUT VIEW

The layout uses a single-sided bottom-copper routing strategy with 0.8 mm power traces (+5V and GND) and 0.5 mm signal routing. Component placement isolates the power entry terminal on the left edge and orients the reed sensor on the perimeter for unimpeded magnetic flux coupling.



5. DELIVERABLE 6: 3D PCB VIEW (KICAD RAYTRACING VISUALIZER)



6. DELIVERABLE 7: COMPLETE BILL OF MATERIALS (BOM)

Item	Designator	Component Description	Value / Specs	Assigned KiCad Footprint	Quantity
1	**J1**	2-Pin Power Screw Terminal	5.0 V DC	`PinHeader_1x02_P2.54mm_Vertical`	1
2	**SW1**	Magnetic Reed Switch Sensor	SPDT / Form-C	`SW_Slide_1P2T_CK_OS102011MS2Q`	1
3	**D1**	Red LED (Door Open Indicator)	5mm, 2.0V V_f	`LED_D5.0mm`	1
4	**D2**	Green LED (Door Closed Indicator)	5mm, 2.2V V_f	`LED_D5.0mm`	1